

Technical Document #984: Data Engineering - Comprehensive Overview

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Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Data pipelines automate the movement and transformation of data between systems.
- Data lineage tracks the origin and movement of data through systems.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse.